§4 理想气体.

§4.1 理想 Fermi 气体

1. $H = \int d\vec{r}\, \psi^+(\vec{r})\, h(\vec{r})\, \psi(\vec{r})$ ，其中 $h(\vec{r}) = -\frac{\hbar^2}{2m}\nabla^2$

$\psi(\vec{r}) = \begin{pmatrix} \psi_{\frac{1}{2}}(\vec{r}) \\ \psi_{-\frac{1}{2}}(\vec{r}) \end{pmatrix}$ ， $\psi^+(\vec{r}) = (\psi_{\frac{1}{2}}(\vec{r})\ \ \psi_{-\frac{1}{2}}(\vec{r}))$

总粒子数巨正则量.

量子统系综： $\rho = \rho(K) = \frac{1}{\Xi} e^{-\beta K}$ ，

其中 $K = H - \mu N$ ， $\Xi = Tr(e^{-\beta K})$ ， $\beta = \frac{1}{k_B T}$

$N = \int d\vec{r}\, \psi^+(\vec{r})\, \psi(\vec{r})$

$N_t = \langle N \rangle = Tr[N\rho(K)]$ 化学势方程.

$J = -k_B T \ln \Xi = -PV$

$P = -\frac{J}{V} = \frac{k_B T}{V} \ln \Xi$

$U = \langle H \rangle = \frac{1}{\Xi} Tr(H e^{-\beta K})$

$S = -k_B \langle \ln \rho(K) \rangle = k_B \beta (\langle H \rangle - \mu \langle N \rangle - J)$

2. 场算符的展开.

$\{u_\alpha(\vec{r})\}$ 为正交归一基底.

$$\begin{cases} \int d\vec{r}\, u_\alpha^*(\vec{r})\, u_{\alpha'}(\vec{r}) = \delta_{\alpha,\alpha'} \\ \sum_\alpha u_\alpha^*(\vec{r})\, u_\alpha(\vec{r}') = \delta(\vec{r}-\vec{r}') \end{cases}$$

$\psi(\vec{r}) = \sum_\alpha \begin{pmatrix} c_{\alpha\uparrow}(\vec{r}) \\ c_{\alpha\downarrow}(\vec{r}) \end{pmatrix} u_\alpha(\vec{r})$ ， $\begin{pmatrix} c_{\alpha\uparrow}(\vec{r}) \\ c_{\alpha\downarrow}(\vec{r}) \end{pmatrix} = \int d\vec{r}\, u_\alpha^*\, \psi(\vec{r})$

$\Rightarrow H = \sum_{\alpha,\alpha',\sigma} c_{\alpha,\sigma}^+ c_{\alpha',\sigma} \int d\vec{r}\, u_\alpha^*(\vec{r})\, h(\vec{r})\, u_{\alpha'}(\vec{r})$ $\sigma = \pm\frac{1}{2}$

$N = \sum_{\alpha,\sigma} c_{\alpha,\sigma}^+ c_{\alpha,\sigma}$

取 $h(\vec{r})\,\eta_\alpha(\vec{r}) = \varepsilon_\alpha \eta_\alpha(\vec{r})$，$h(\vec{r}) = -\frac{\hbar^2}{2m}\nabla^2$

则 $H = \sum_{\alpha,\sigma} \varepsilon_\alpha C^+_{\alpha,\sigma} C_{\alpha,\sigma}$

$K = \sum_{\alpha,\sigma} (\varepsilon_\alpha - \mu) C^+_{\alpha,\sigma} C_{\alpha,\sigma}$

取周期边界条件

$\alpha = \vec{k}$，

$k_1 = \frac{2\pi}{L_1} n_1$，$k_2 = \frac{2\pi}{L_2} n_2$，$k_3 = \frac{2\pi}{L_3} n_3$

$\eta_{\vec{k}} = \frac{1}{\sqrt{\Omega}} e^{i\vec{k}\cdot\vec{r}}$，$\varepsilon_{\vec{k}} = \frac{\hbar^2 k^2}{2m}$

则 $H = \sum_{\vec{k},\sigma} \varepsilon_{\vec{k}} C^+_{\vec{k},\sigma} C_{\vec{k},\sigma}$，

$N = \sum_{\vec{k},\sigma} C^+_{\vec{k},\sigma} C_{\vec{k},\sigma}$

$K = \sum_{\vec{k},\sigma} (\varepsilon_{\vec{k}} - \mu) C^+_{\vec{k},\sigma} C_{\vec{k},\sigma}$

$H = \sum_{\vec{p},\sigma} V_{\vec{p},\sigma}$

$Z = \mathrm{Tr}\, e^{-\beta \sum_{\vec{k},\sigma} (\varepsilon_{\vec{k}} - \mu) n_{\vec{k},\sigma}}$

$= \mathrm{Tr} \left(\prod_{\vec{k},\sigma} e^{-\beta(\varepsilon_{\vec{k}} - \mu) n_{\vec{k},\sigma}} \right)$

$= \prod_{\vec{k},\sigma} \mathrm{Tr}_{\vec{k},\sigma} \left(e^{-\beta(\varepsilon_{\vec{k}} - \mu) n_{\vec{k},\sigma}} \right)$

$= \prod_{\vec{k},\sigma} \sum_{n_{\vec{k},\sigma}=0}^{1} e^{-\beta(\varepsilon_{\vec{k}} - \mu) n_{\vec{k},\sigma}}$

$= \prod_{\vec{k},\sigma} \left[1 + e^{-\beta(\varepsilon_{\vec{k}} - \mu)} \right]$

$= [$

$$\langle n_{\vec{R},\sigma} \rangle = \frac{1}{Z} \mathrm{Tr}\left\{ n_{\vec{R},\sigma} e^{-\beta \sum_{\vec{R}',\sigma'} (\varepsilon_{\vec{R}'} - \mu) n_{\vec{R}',\sigma'}} \right\}$$

$$= \frac{1}{Z} \mathrm{Tr}\left\{ n_{\vec{R},\sigma} \prod_{\vec{R}',\sigma'} e^{-\beta(\varepsilon_{\vec{R}'} - \mu) n_{\vec{R}',\sigma'}} \right\}$$

4. 态密度

$$G := \frac{1}{V}\sum_{k,\sigma} g(\varepsilon_k)$$

$$= \frac{1}{V}\sum_{k,\sigma}\int_{-\infty}^{\infty} d\omega\, \delta(\omega-\varepsilon_k)\, g(\omega)$$

$$= \int_{-\infty}^{\infty} d\omega \left(\frac{1}{V}\sum_{k,\sigma}\delta(\omega-\varepsilon_k)\right) g(\omega)$$

$$= \int_{-\infty}^{\infty} d\omega\, N(\omega)\, g(\omega)$$

态密度 $N(\omega) = \frac{1}{V}\sum_{k,\sigma}\delta(\omega-\varepsilon_k)$

热力学极限

态密度 $D(\omega)$

$\delta(x) = \frac{d}{dx}\theta(x)$, $\theta(x) = \begin{cases} 1 & x>0 \\ 0 & x<0 \end{cases}$

$$D(\omega) = \lim_{V\to\infty}\frac{d}{d\omega}\frac{1}{V}\sum_{k,\sigma}\theta(\omega-\varepsilon_k)$$

$$= \lim_{V\to\infty}\frac{d}{d\omega}\left[\frac{1}{V}\sum_{k,\sigma}\theta(\omega-\varepsilon_k)\right]$$

$$= \frac{d}{d\omega}\lim_{V\to\infty}\left[\frac{1}{V}\sum_{k,\sigma}\theta(\omega-\varepsilon_k)\right]$$

$\vec{k}$ 空间中一个点占的体积 $= \frac{2\pi}{L_x}\cdot\frac{2\pi}{L_y}\cdot\frac{2\pi}{L_z} = \frac{(2\pi)^3}{V}$

$$\lim_{V\to\infty}\left[\frac{1}{V}\sum_{k,\sigma}\theta(\omega-\varepsilon_k)\right] = 2\lim_{\Delta\vec{k}\to 0}\sum_{\vec{k}}\frac{(2\pi)^3}{V}\theta(\omega-\varepsilon_k)$$

$$= 2\int_{\mathbb{R}^3} d\vec{k}\,\frac{1}{(2\pi)^3}\,\theta(\omega-\varepsilon_k) = 2\int_0^{+\infty}\frac{1}{(2\pi)^3}\,4\pi k^2\,dk\,\theta(\omega-\varepsilon_k)$$

$$= \frac{1}{2\pi^2}\int_0^{+\infty} k^2\,\theta\left(\omega-\frac{\hbar^2k^2}{2m}\right)dk = \frac{1}{2\pi^2}\int_0^{+\infty}\frac{2m}{\hbar^2}\sqrt{\frac{\hbar^2}{2m}}\sqrt{\omega}\,\theta(\omega-\varepsilon_k)\,d\varepsilon_k$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$= \frac{1}{2\pi} \left(\frac{2m}{\hbar^2}\right)^{3/2}$$

§ 4.2 理想 Boson 系统.

1. 同 4.1.1

2. 粒子数不定.

$\psi(\vec{r}) = \frac{1}{\sqrt{V}} \sum_{\vec{k}} a_{\vec{k}} e^{i\vec{k}\cdot\vec{r}}$, $\psi^{+}(\vec{r}) = \frac{1}{\sqrt{V}} \sum_{\vec{k}} a_{\vec{k}}^{+} e^{-i\vec{k}\cdot\vec{r}}$